

VxLEARN Networks

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report: The Client Interaction

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1. Introduction

This lab demonstrates how a client PC interacts with a server when requesting a web page.

Packet Tracer simulates the complete exchange:

- DNS lookup (resolving www.example.com to an IP)
- HTTP request sent from PC to server
- HTTP response sent back to PC
- TCP acknowledgements
- Segmented web data transfer

Understanding these interactions is fundamental in networking because nearly all modern communication relies on these two services: DNS to find the server, and HTTP to retrieve content.

2. Network Topology (Before Simulation)

The network is extremely simple:

- One PC
- One Server
- Directly connected with a copper cable
- The server runs DNS and HTTP services
- The PC uses the server for DNS resolution

This model helps isolate and observe exactly how URL resolution and webpage loading occur.



Figure 1: The simple network topology for client–server interaction demonstration.

3. Part 1 – Enter Simulation Mode

Simulation Mode allows Packet Tracer to show the **exact packets, protocols, and Layer-by-Layer OSI processing**.

Steps:

1. Click **Simulation Mode** in the bottom-right corner.
2. The event-list panel becomes visible.
3. Packet events will now pause and wait for manual forwarding.



Figure 2: Simulation mode activated, showing the Event List panel.

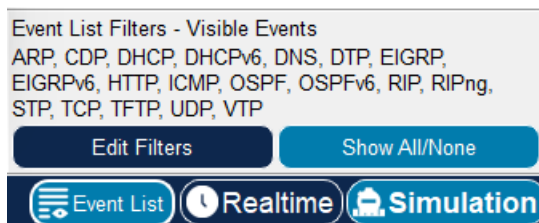
4. Part 2 - Set Event List Filters

The default setting shows **all** packet types, which overwhelms the screen.
We only want:

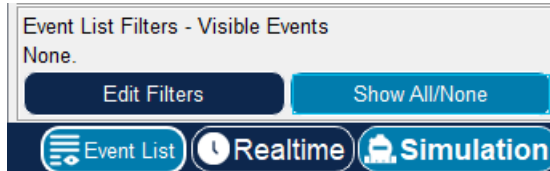
- **DNS packets**
- **HTTP packets**

Steps:

1. Click **Show All/None** → **None**, clearing all protocols.

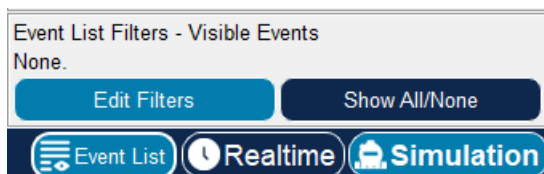


Before

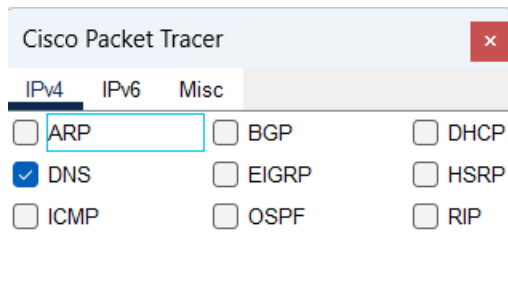


After

2. Click **Edit Filters**.



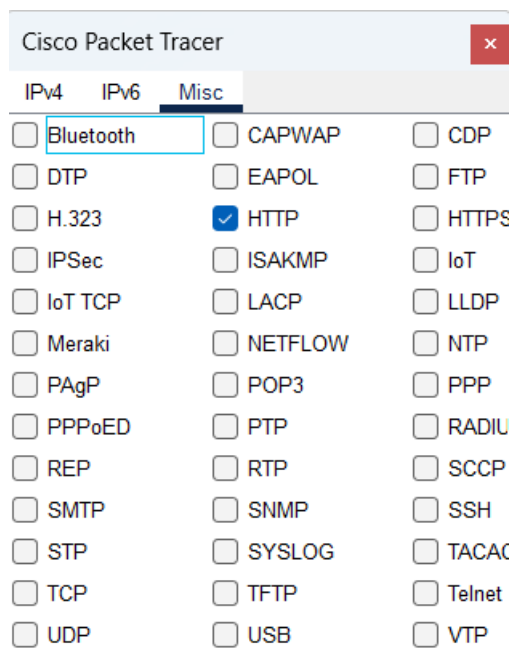
3. Under **IPv4**, select **DNS**.



The screenshot shows the 'Cisco Packet Tracer' window with the 'IPv4' tab selected. The 'Misc' sub-tab is active, displaying a list of protocols. The 'DNS' checkbox is checked, while others like ARP, BGP, DHCP, EIGRP, HSRP, ICMP, OSPF, and RIP are unchecked.

Protocol	Selected
ARP	☐
DNS	☒
ICMP	☐
BGP	☐
EIGRP	☐
OSPF	☐
DHCP	☐
HSRP	☐
RIP	☐

4. Under **Misc**, select **HTTP**.



The screenshot shows the 'Cisco Packet Tracer' window with the 'Misc' tab selected. The 'HTTP' checkbox is checked, while others like Bluetooth, CAPWAP, CDP, DTP, EAPOL, FTP, H.323, HTTPS, IPSec, IoT, LLDP, Meraki, NTP, PAgP, POP3, PPP, PPPoED, RADIUS, REP, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, and VTP are unchecked.

Protocol	Selected
Bluetooth	☐
DTP	☐
H.323	☐
IPSec	☐
IoT TCP	☐
Meraki	☐
PAgP	☐
PPPoED	☐
REP	☐
SMTP	☐
STP	☐
TCP	☐
UDP	☐
CAPWAP	☐
EAPOL	☐
ISAKMP	☐
LACP	☐
NETFLOW	☐
POP3	☐
PTP	☐
RTP	☐
SNMP	☐
SYSLOG	☐
TFTP	☐
USB	☐
CDP	☐
FTP	☐
HTTPS	☐
IoT	☐
LLDP	☐
NTP	☐
PPP	☐
RADIUS	☐
SCCP	☐
SSH	☐
TACACS	☐
Telnet	☐
VTP	☐
HTTP	☒

5. Close.

Now you will only see the important packets related to the client's web activity.

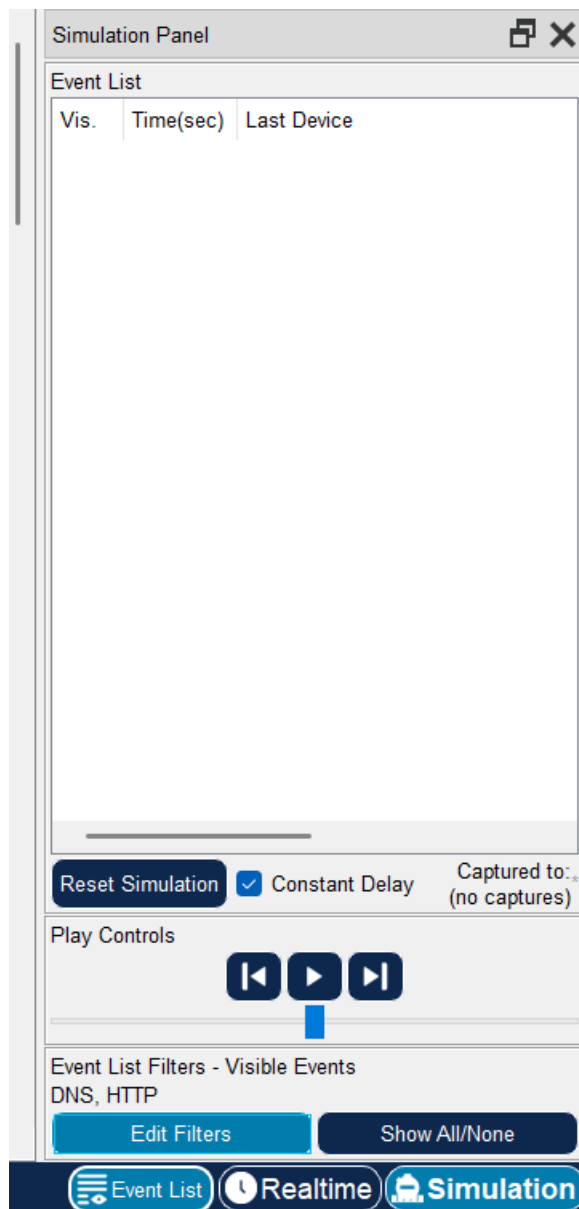


Figure 3: Event List Filters configured to display only DNS and HTTP packets.

5. Part 3 – Requesting a Web Page

Steps:

1. Click **PC** → **Desktop** → **Web Browser**.
2. Enter:

`www.example.com`

3. Click **Go**.
4. Minimize (do not close) the browser.

This triggers:

- A DNS lookup
- Then an HTTP GET request

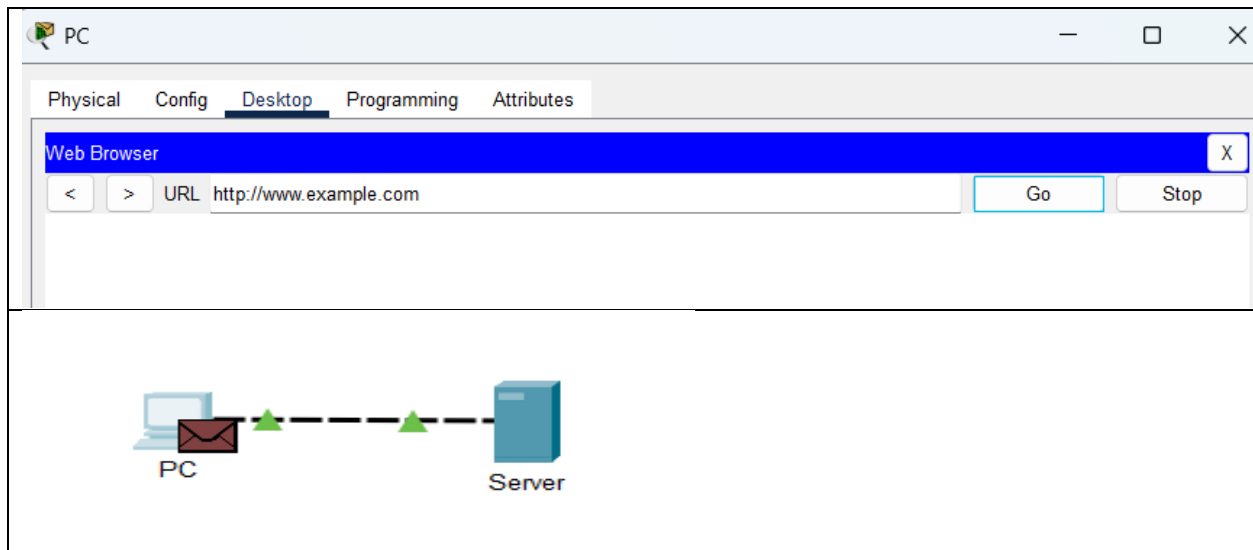
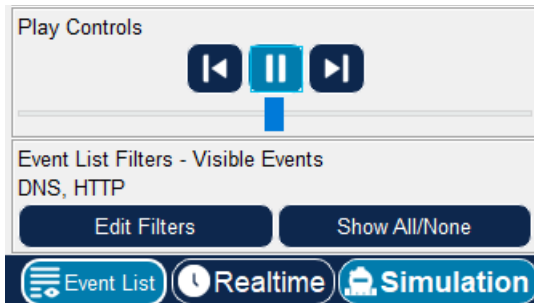


Figure 4: Simulated PC submitting website request to www.example.com

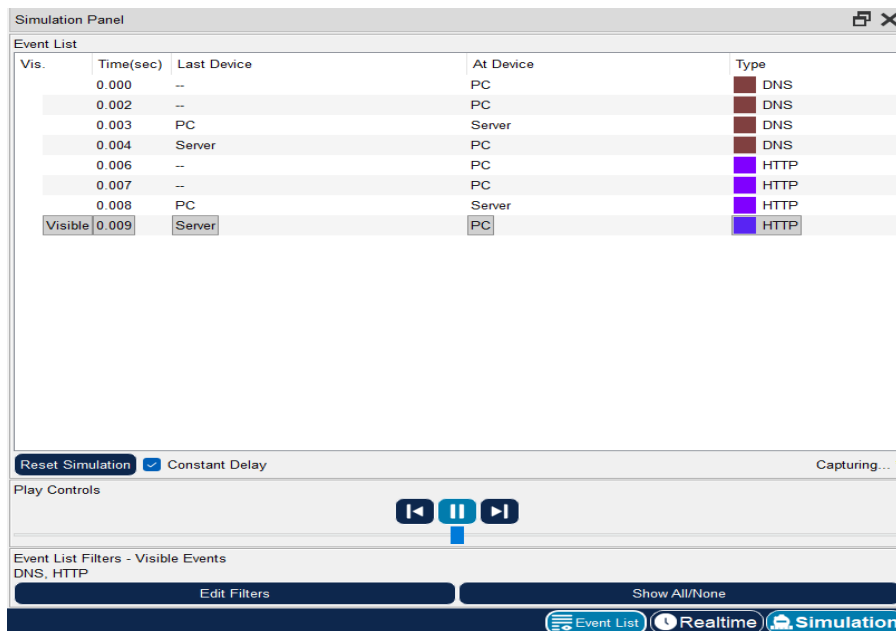
6. Part 4 – Run the Simulation

Steps:

1. In Simulation Panel, click **Play**.



2. Watch the packets flow:
 - DNS query → server
 - DNS response → PC
 - HTTP GET → server
 - HTTP DATA (segments 1 and 2)
 - TCP ACKs
3. If the buffer fills, click **View Previous Events**.





This models an actual, real-life client–server communication.

7. Part 5 – Inspecting the PDU

Now we inspect each exchange at the OSI layer level.

Steps:

1. In Event List, click the **colored square** next to the first event (DNS request).

Simulation Panel				
Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC	 DNS

2. The **PDU Information window** opens.
3. Click **Next Layer >>** repeatedly to observe how each OSI layer processes the packet.

PDU Information at Device: PC

OSI Model

Outbound PDU Details

At Device: PC
Source: PC
Destination: 192.168.1.2

In Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer2
Layer1

Out Layers
Layer 7: DNS
Layer6
Layer5
Layer 4: UDP Src Port: 1025, Dst Port: 53
Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.2
Layer 2:
Layer1

1. The DNS client sends an A DNS query to the DNS server.

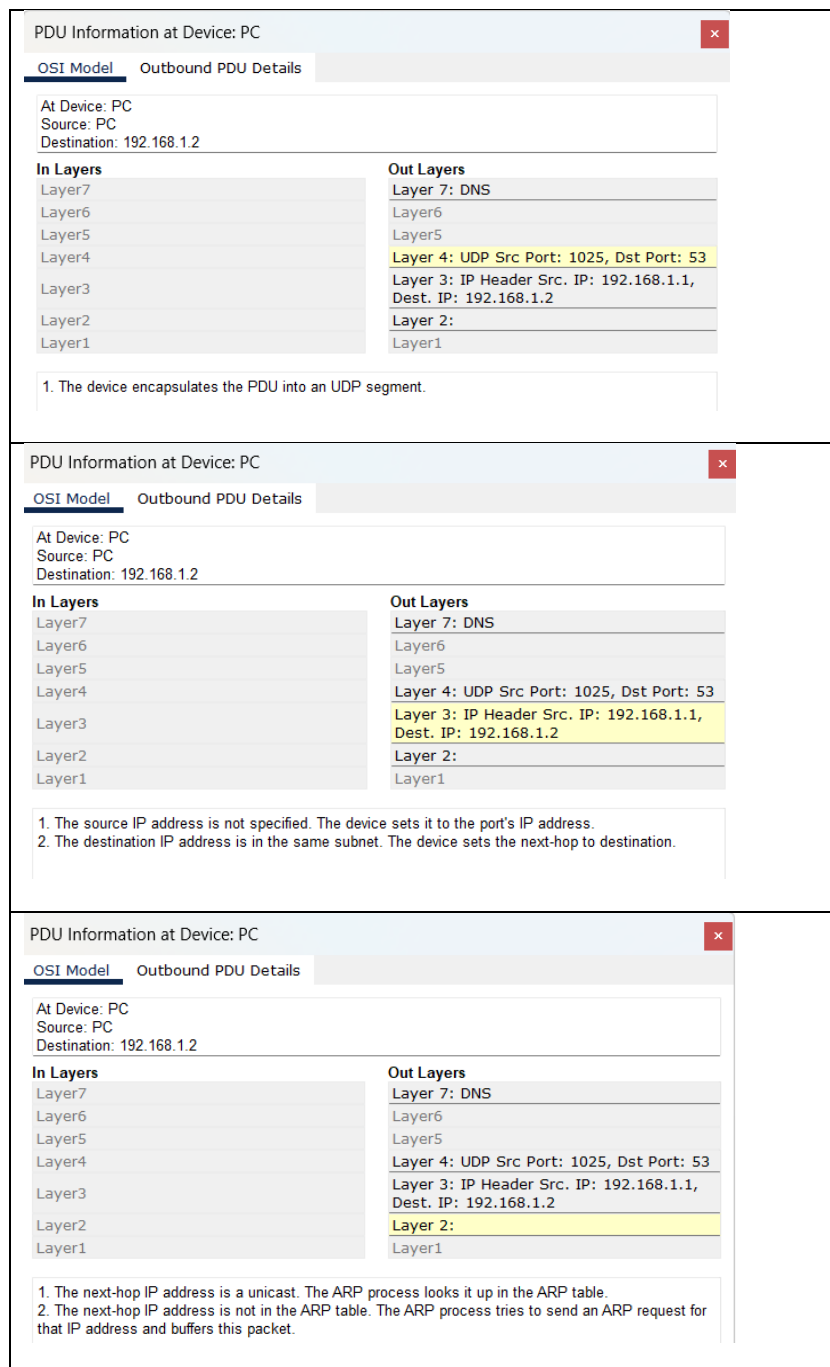


Figure 5: Detailed OSI-layer view of the DNS request packet.

8. Part 6 – Understanding the Client Interaction

DNS Step

When the PC enters www.example.com, it does not know the IP address.

So it sends a:

DNS QUERY: “What is the IP of www.example.com?”

The server replies:

DNS RESPONSE: “The IP is 10.0.0.10”

Only after this can the PC contact the web server.

HTTP Step

Once the PC knows the IP, it sends:

HTTP GET /index.html

The server responds by splitting the webpage into segments:

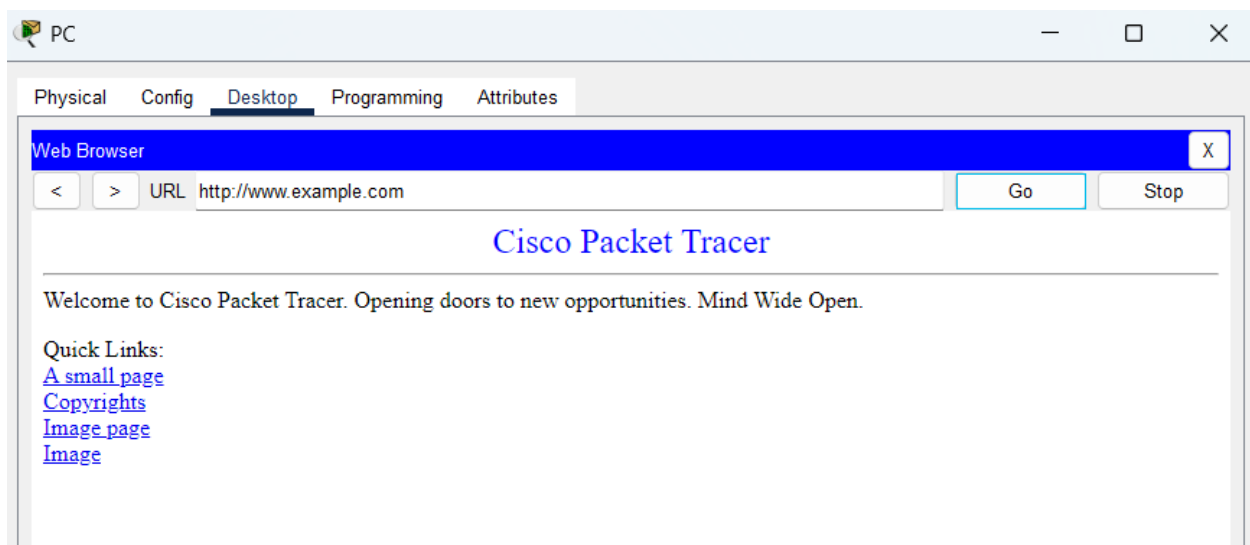
HTTP DATA segment 1

HTTP DATA segment 2

The PC acknowledges each:

TCP ACK

Then the webpage is displayed on the browser.



9. Summary & Reflection

What I learned:

- DNS converts domain names to IP addresses (critical for internet use).
- HTTP retrieves webpage data using segmented TCP delivery.
- OSI layers control how data is packaged, transmitted, routed, received, and displayed.
- Packet Tracer provides an excellent view of real packet flow in a simplified environment.

Why this matters:

Every time you visit a website in real life, your device performs the exact steps seen in this simulation DNS, HTTP GET, segmentation, acknowledgements, and webpage rendering.